Innovation & Entrepreneurship for Engineers

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IIT Madras

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Innovation & Entrepreneurship for Engineers

Module 1
Engineers as Innovators

#1. What is Innovation ?
#2. Innovation through S & T

Module 2
Problem Discovery

#3. Finding Problems worth solving
#4. Customer Discovery & Design Thinking

Module 3
Creativity & Innovation

#5.Innovative Thinking
#6. Enhancing Innovation – AI - Collaboration

Module 4
Deep-Tech Entrepreneurship

#7. Deep – Tech Ventures
#8. Exponential & Systems Thinking

Module 5
Building a Startup

#9. Economics of Innovation & Entrepreneurship
#10. Business Models
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Module 6
Prototyping & Presentation

#12. Rapid Prototyping
#13. Pitch Your Idea
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Innovation & Entrepreneurship for Engineers

Deep Tech Ventures

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The Deep-Tech Venture Journey

1 • DISCOVER
What makes a venture deep tech?

2 • COMMERCIALISE
How does research escape the lab?

3 • PROVE
How does technology mature?

4 • DE-RISK & SCALE
Why are deep-tech ventures hard?

SCIENCE → TECHNOLOGY → PRODUCT → VENTURE → IMPACT

Learning outcomes

- Distinguish deep-tech, deep-engineering and technology-enabled ventures.
- Map research to customer value and commercialization.
- Use NASA-style TRL thinking with evidence.
- Identify technical, resource, regulatory, repeatability and revenue risks.
- Design the next experiment that reduces the most important uncertainty.
- Assess a real Indian deep-tech venture and make a defendable investment recommendation.

Opening question

What is the difference between a startup that USES technology and one whose competitive advantage IS the technology?

Technology-enabled
Technology enables a new business model.

Deep engineering
Difficult systems integration creates the moat.

Science-based deep tech
Frontier science / technology is central.

Deep tech is a spectrum — not a binary label.

DISCOVER

What makes a venture deep tech?

The 4D Deep-Tech Test

D1 • DISCOVERY
Scientific or advanced engineering knowledge is central.

D2 • DIFFICULT
The problem cannot be solved by simply assembling existing technology.

D3 • DEFENSIBLE
IP, know-how, data or manufacturing capability creates a moat.

D4 • DEVELOPMENT
Substantial experimentation, engineering, capital or time is required.

Which of the four Ds makes THIS venture difficult to copy?

Six Pathways

Case	Primary pathway	Teaching question
Agnikul Cosmos	Advanced engineering	Can we reinvent how rockets are designed/manufactured?
Ather Energy	Systems engineering	Can integration itself become a technology moat?
Sarvam AI	Frontier AI platform	Can a national problem create a platform technology?
SpaceX	Breakthrough engineering	What accepted assumption can technology overturn?
DeepMind	Fundamental research	Can research itself become a platform?
Moderna	Scientific platform	Can one scientific platform generate many products?

Case lens: Agnikul Cosmos

Advanced engineering + Propulsion + Additive manufacturing

Core problem: access to space for small satellites and missions.

Technology lens: propulsion, additive manufacturing, avionics and launch systems.

Lesson: the opportunity is to rethink design, manufacturing and operations as a system.

Deep-tech entrepreneurship = difficult technology + valuable problem + defensibility + scale.

Case lens: Ather Energy

Deep engineering + systems integration + productization

Useful as an engineering-intensive / deep-engineering case rather than a pure research spinout.

Technology stack: battery / BMS, motor, power electronics, controls, software, connectivity and charging.

Lesson: sophisticated integration and accumulated engineering know-how can become a moat

Deep-tech entrepreneurship = difficult technology + valuable problem + defensibility + scale.

Case lens: Sarvam AI

Frontier AI + Indian-language capability + platform

Core technology: foundation models, speech, language and AI infrastructure.

Moat can combine models, data, compute, engineering, deployment and ecosystem.

Question: model, platform, applications—or all three?

Deep-tech entrepreneurship = difficult technology + valuable problem + defensibility + scale.

Case lens: SpaceX

Breakthrough engineering + manufacturing + system economics

Accepted assumption challenged: launch vehicles are largely expendable.

Technology stack: propulsion, materials, software, manufacturing, operations and reusability.

Lesson: technical improvement can change industry economics.

Deep-tech entrepreneurship = difficult technology + valuable problem + defensibility + scale.

Case lens: DeepMind

Fundamental research → platform technology → applications

Research-intensive AI is the starting point.

Value can come from pushing the frontier of knowledge itself.

Lesson: research capability can repeatedly generate valuable applications.

Deep-tech entrepreneurship = difficult technology + valuable problem + defensibility + scale.

Case lens: Moderna

Scientific platform → product → platform expansion

Core idea: mRNA as a programmable therapeutic / vaccine platform.

Lesson: the real asset may be a platform that can generate many future products.

Question: what is the platform—and what is one application?

Deep-tech entrepreneurship = difficult technology + valuable problem + defensibility + scale.

The Deep-Tech Venture Equation

BREAKTHROUGH
What is scientifically / technically new?

PROBLEM
What important problem does it solve?

PLATFORM
Can the technology serve multiple applications?

MOAT
Why is it difficult to copy?

SCALE
Can it become economically significant?

Deep-Tech Venture = Breakthrough × Problem × Platform × Moat × Scale

Tool: Deep-Tech Venture Canvas

Question	Team answer
Scientific / engineering breakthrough	
Important problem	
Who experiences it?	
Technology enabling the solution	
IP / know-how / data moat	
Other applications	
What must be proven technically?	
What must be proven commercially?	
What could kill the venture?	
What could make it enormous?	

2 • COMMERCIALISE

How does research escape the lab?

The Research-to-Impact Gap

DISCOVERY → INVENTION → TECHNOLOGY → PRODUCT → BUSINESS → IMPACT

A paper creates knowledge.

A patent protects an invention.

A prototype demonstrates feasibility.

A product creates customer value.

A business creates economic value

The hardest bridge is often INVENTION → COMMERCIALISATION.

Framework: D-P-P-T-P-S

DISCOVER
What capability exists?

PROTECT
What can be protected?

PROVE
Can it actually work?

TRANSLATE
How does it create customer value?

PILOT
Can it work in reality?

SCALE
Can it be repeated economically?

The questions change as technology matures

Stage	Primary question
Research	Is it true?
Engineering	Can we make it work?
Entrepreneurship	Does anyone value it enough to pay?
Operations	Can we deliver it repeatedly?
Scale	Can it become economically significant?

Tool: Capability → Problem → Customer → Value

Technology capability	Possible problem	Customer	Measurable value
High temperature			
Lightweight			
High conductivity			
Self-healing			
High sensitivity			

Research-to-Market Canvas

Block	Question
SCIENCE	What scientific principle has been discovered?
CAPABILITY	What can the technology now do?
PROBLEM	Which important problem could it solve?
CUSTOMER	Who suffers from the problem?
VALUE	What measurable benefit is created?
DIFFERENTIATION	Why is it better than alternatives?
IP	What can be protected?
TRL	How mature is it?
BEACHHEAD	What is the first application?
SCALE	What prevents mass deployment?
NEXT EXPERIMENT	What must we learn next?

The 6 GATE Commercialization Test

1 • GOOD SCIENCE
Does the phenomenon work?

2 • GENUINE PROBLEM
Does somebody care?

3 • GREAT ADVANTAGE
Is improvement meaningful?

4 • GUARDABLE
Can defensibility be created?

5 • AFFORDABLE
Can economics work?

6 • EXPANDABLE
Can it become more than one product?

Commercialize the VALUE created by the technology — not the technology itself.

Mini-exercise • Research Commercialisation

Imagine: "We have developed a material that withstands 500°C higher temperature than current materials."

Generate three application spaces.

For each: problem, customer and measurable value.

Choose one beachhead.

State the evidence needed before forming a company

Ask "What important problem does this capability unlock?"

3 • PROVE

How does technology mature?

What is TRL?

NASA's Technology Readiness Levels are a nine-level scale for assessing technology maturity.

TRL 1 = basic principles observed and reported.

TRL 9 = actual system successfully operated in the operational environment.

Assign a TRL from evidence against criteria — not optimism

TRL measures TECHNOLOGY MATURITY.
It does not automatically measure MARKET ATTRACTIVENESS.

NASA TRL 1–9 at a glance

TRL	Meaning
1	Basic principles observed and reported
2	Technology concept / application formulated
3	Critical function / characteristic proof of concept
4	Component / breadboard validated in laboratory
5	Component / breadboard validated in relevant environment
6	System / subsystem model or prototype demonstrated in relevant environment
7	System prototype demonstrated in operational / space environment
8	Actual system completed and qualified
9	Actual system proven through successful mission / operational use

Running example: Solid-State Battery for an EV

TRL	What must be demonstrated?
1	Solid electrolyte exhibits relevant ion transport
2	Concept: replace liquid electrolyte in a battery
3	Lab cell demonstrates critical electrochemical function
4	Integrated cell components validated in lab
5	Cell tested under representative temperature/load conditions
6	Multi-cell/module prototype demonstrated in relevant environment
7	Pack-level system demonstrated in vehicle-relevant operation
8	Production-intent pack qualified against requirements
9	Commercial vehicle operates successfully with the technology

The evidence ladder

WHAT?
What exactly was demonstrated?

WHERE?
In what environment?

HOW?
What test / data / demonstration?

SO WHAT?
What decision does evidence enable?

“We have a prototype” is not a TRL assessment. “Here is the evidence” is.

TRL Evidence Card

Field	Answer
Current TRL	
What has been demonstrated?	
What evidence exists?	
Environment tested	
Critical parameter proven	
What remains unknown?	
Experiment to reach next TRL	
Exit criterion	
Time / cost to evidence	

Technology Readiness ≠ Venture Readiness

T • TECHNOLOGY

Can we make it work?

M • MARKET

Does anyone want it?

O • OPERATIONS

Can we deliver it repeatedly and economically?

A technology can be TRL 9 and still be a terrible business !

TRL → Funding → Milestones

Stage	Primary objective	Investor question
TRL 1–2	Scientific feasibility	Is the principle worth pursuing?
TRL 3–4	Technical feasibility	Can we build it?
TRL 5–6	Technology demonstration	Does it work in relevant conditions?
TRL 7–8	Product qualification	Can we make a reliable product?
TRL 9	Operational use	Does it work repeatedly in the field?

Mini-exercise • TRL Detective

- Choose a technology claim from a startup, lab or news article.
- Find the evidence behind the claim.
- Map it to NASA’s TRL definition.
- Identify the next evidence gate.
- State what would make you revise your assessment

Don’t tell me your technology is ready. Show me the evidence.

4 • DE-RISK & SCALE

Why are deep-tech ventures hard?

Why deep-tech ventures are different

Longer time between discovery and commercialization.

Higher technical uncertainty.

More expensive experiments and specialized infrastructure.

Complex regulation, certification and safety requirements.

Harder transition from prototype to reliable manufacturing/deployment.

Customer adoption may require changing existing workflows or systems.

Capital must often arrive before conventional revenue signals appear

Reduce the uncertainties that matter most.

Framework: The 5R Deep-Tech Challenge Map

R1 • RESEARCH RISK
Will the science / technology work?

R2 • RESOURCE RISK
Can we obtain capital, talent, facilities, partners and time?

R3 • REGULATORY RISK
Can we legally certify and deploy it?

R4 • REPEATABILITY RISK
Can we manufacture / deploy reliably?

R5 • REVENUE RISK
Can the technology generate attractive economics?

The Deep-Tech Risk Matrix

	LOW UNCERTAINTY	HIGH UNCERTAINTY
LOW IMPACT	MONITOR	EXPERIMENT
HIGH IMPACT	EXECUTE	KILLER RISK

The most important risk is not necessarily the most visible risk.

The Deep-Tech Assumption Stack

SCIENTIFIC
Does the physics / chemistry / biology work?

TECHNICAL
Can we engineer it?

MANUFACTURING
Can we produce it?

CUSTOMER
Will customers adopt it?

ECONOMIC
Will unit economics work?

REGULATORY
Can it be approved?

SCALING
Can it grow 100×?

Tool: Assumption → Experiment → Evidence

Assumption	Test / experiment	Success signal	Failure signal

Prioritise experiments by information gained per unit time and money.

From Scientist → Engineer → Entrepreneur → Leader

Role	Core question
Scientist	Is it possible?
Engineer	Can we make it work?
Entrepreneur	Does it create enough value?
Business leader	Can we scale it?
Societal leader	Should we scale it?

Deep-tech leadership connects scientific truth, engineering reality, customer value and societal impact.

Four-session Synthesis

DISCOVER
4D Deep-Tech Test

COMMERCIALISE
D-P-P-T-P-S

PROVE
TRL + evidence gates

DE-RISK
5R + assumption tests

DISCOVER → COMMERCIALISE → PROVE → DE-RISK → SCALE

ASSIGNMENT

Assignment objective

How has—or how could—advanced science and engineering become a valuable, defensible and scalable venture?

Show what the technology does.

Explain why it is difficult to develop.

Show how it creates customer value.

Assess the evidence supporting readiness.

Identify major risks and unknowns.

Recommend what the company should test or achieve next.

The four cases

Company	Domain	Core analytical question
GalaxEye	Space / Earth observation	Can sensor fusion create a better EO platform and scalable business?
Qure.ai	AI / healthcare	How does advanced AI become clinically useful, trusted and regulated?
Planys Technologies	Marine robotics / NDT	How does research in robotics become a valuable industrial service?
Mindgrove Technologies	Semiconductors / SoCs	Why is building a semiconductor venture different from a software startup?

Case A • GalaxEye

Space technology + Earth observation

Founded in 2021 by engineers from IIT Madras.

Mission Drishti combines SAR and multispectral optical sensing on one satellite platform.

The company positions OptoSAR as combining optical clarity with SAR's all-weather/day-night capability.

Investigate technology, satellite architecture, data product, customers and path to scale

Separate COMPANY FACTS from TEAM INFERENCES.

Case B • Qure.ai

AI + healthcare + medical-device regulation

Founded in 2016 with a vision to make healthcare more accessible through AI.

Uses deep learning and AI across areas including TB, lung cancer and stroke.

Multiple regulatory clearances make evidence and regulation part of the commercialization story.

Investigate clinical evidence, workflow integration, regulatory readiness, adoption and economics

Separate COMPANY FACTS from TEAM INFERENCES.

Case C • Planys Technologies

Marine robotics + NDT + industrial inspection

Emerges from IIT Madras research/student work and develops underwater inspection solutions.

Technology combines marine robotics, sensors, NDT and analytics.

Applications include inspection of difficult or hazardous infrastructure.

Investigate research origin, customer problem, service model, deployment evidence and scaling

Separate COMPANY FACTS from TEAM INFERENCES.

Case D • Mindgrove Technologies

Semiconductors + SoCs + embedded systems

Fabless semiconductor design company.

SoCs and platforms target applications including IoT, automotive, industrial automation, vision and edge AI.

The ecosystem spans architecture, verification, fabrication, packaging, testing and software.

Investigate what the company controls and what the ecosystem must provide

Separate COMPANY FACTS from TEAM INFERENCES.

Assignment Task 1 • Why is it deep tech?

Dimension	Question
Discovery	What scientific / advanced engineering knowledge is central?
Difficult	What makes the problem technically difficult?
Defensible	What could make it difficult to copy?
Development	Why is substantial experimentation / engineering / capital needed?

Conclusion: Why is this company deep tech rather than simply technology-enabled?

Assignment Task 2 • Research → Market

Stage	Evidence to find
Discover	Scientific / engineering capability
Protect	IP / know-how / data / ecosystem advantage
Prove	First demonstration
Translate	Technology → customer value
Pilot	Real-world deployment
Scale	Repeatability + economics

Assignment Task 3 • Find the best application

Technology capability	Problem	Customer	Value

Distinguish current company applications from applications invented by your team.

Assignment Task 4 • Build the TRL journey

Map the technology from TRL 1–9 where meaningful.

Do not assign a number without evidence.

Identify current TRL and supporting evidence.

Identify next TRL and exit criterion.

Propose the experiment/test that creates the missing evidence

The question is not “What TRL are you?” It is “What evidence earns the next TRL?”

Assignment Task 5 • 3D Readiness

Dimension	Question	Evidence
Technology readiness	Can it work?	
Market readiness	Does anyone want it?	
Operational / manufacturing readiness	Can it be delivered repeatedly and economically?	

Which readiness dimension is most likely to constrain growth?

Assignment Task 6 • 5R Risk Map

Risk	Question
Research	Will the science / technology work?
Resources	Can we obtain capital, talent, equipment, partners and time?
Regulation	Can we certify and deploy?
Repeatability	Can we manufacture / deploy reliably?
Revenue	Can the technology create attractive economics?

Identify 3 major risks + 1 potential KILLER RISK.

Assignment Task 7 • Design the next experiments

Assumption	Experiment	Success signal	Failure signal

Choose the experiment that creates the most learning for the least time and money.

Assignment Task 8 • Deep-Tech Venture Canvas

Element	Question
Technology	What is technically distinctive?
Problem	What important problem is being solved?
Customer	Who experiences it?
Value	What measurable value is created?
Moat	Why is it difficult to copy?
Beachhead	What is the first market/application?
TRL	What is the current maturity?
Market readiness	What evidence shows demand?
Scale	What blocks 100× scale?
Next milestone	What must be demonstrated next?

Company-specific questions

Use these in addition to the common framework

GALAXEYE

Optical + SAR integration; EO customers; satellite/data/platform; scale.

QURE.AI

Clinical evidence; regulation; workflow; customer/payer; AI moat.

PLANYS

Research origin; underwater inspection; service model; deployment; scale.

MINDGROVE

SoC lifecycle; fabless ecosystem; scarce resources; defensibility; capital/time.

Final Team Challenge • ₹10 crore / 24 months

Choose 3 technical milestones.

Choose 2 customer / market experiments.

Choose 1 manufacturing / scale experiment.

State the evidence that must exist after 24 months to justify further investment.

Do not allocate the entire budget to “more R&D.

Capital should buy milestones and evidence—not simply more time.

The “Kill the Venture” Test

Strongest reason NOT to invest: _____

Most dangerous assumption: _____

Easiest important assumption to test: _____

Evidence most likely to change our mind: _____

If the assumption fails: PIVOT / MODIFY / PAUSE / QUI

A good deep-tech team knows what would prove them wrong.

Submission: 6-slide Deep-Tech Venture Analysis

Slide	What to show
1 • The Venture	Technology + problem + customer
2 • Why Deep Tech?	4D test + conclusion
3 • Research → Market	D-P-P-T-P-S + evidence
4 • Technology Readiness	Current TRL + evidence + next TRL
5 • Major Risks	5R map + killer assumption + experiment
6 • Should We Bet?	Double down / modify / watch / do not bet

Presentation & cross-examination

5 minutes + 3 minutes

SCIENTIST
Challenge scientific assumptions.

ENGINEER
Challenge technical evidence.

CUSTOMER
Challenge value and adoption.

INVESTOR
Challenge economics and risk.

REGULATOR
Challenge safety, standards and approval.

Every team member should defend the venture from one perspective.

Assessment rubric

100 points

Criterion	Weight
Deep-tech character	15%
Research commercialisation	15%
TRL + evidence analysis	20%
Risks + unknowns	20%
Experiments + evidence	15%
Commercial / venture thinking	10%
Clarity / presentation	5%

Reward reasoning from evidence—not admiration for a successful company.

One-page team summary

1. The company is solving…
2. Its deep-tech advantage is…
3. The underlying scientific/engineering breakthrough is…
4. Its strongest potential moat is…
5. Its current technology readiness is…
6. The evidence supporting this assessment is…
7. Its biggest market uncertainty is…
8. Its biggest technical uncertainty is…
9. The next experiment we would conduct is…
10. We would invest / not invest because…

The central question

“What must be true for this deep-tech venture to become a great company—and what evidence do we need to find out?”

DISCOVER → PROVE → DE-RISK → SCALE

A breakthrough creates possibility.

Evidence creates confidence.

Entrepreneurship creates value.

Scale creates impact.

Sources & teaching discipline

Official sources should be the default for current company facts

NASA — Technology Readiness Levels: nine-level maturity scale and evidence/exit criteria.

GalaxEye — About / Mission Drishti: company history, OptoSAR architecture and mission information.

Qure.ai — About / Regulatory / Evidence: products, clearances and clinical evidence.

IIT Madras / Research Park — deep-tech ecosystem and venture context.

For Planys and Mindgrove: verify current product/technology details from company and IIT Madras sources before class.

Throughout the assignment, label statements as FACT • INFERENCE • HYPOTHESIS • UNKNOWN

NASA TRL reference:
https://www.nasa.gov/directorates/somd/space-communications-navigation-program/technology-readiness-levels/

The Deep-Tech Venture Journey

1 • DISCOVER
What makes a venture deep tech?

2 • COMMERCIALISE
How does research escape the lab?

3 • PROVE
How does technology mature?

4 • DE-RISK & SCALE
Why are deep-tech ventures hard?

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